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The Welfare Effects of Peer Entry: The Case of Airbnb and the Accommodation Industry

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Markets, products, and sample	3
2.2	Data sources	4
2.3	Main empirical objects	4
3	Reduced-form evidence (all regressions + interpretation)	5
3.1	City-level predictors of Airbnb penetration	5
3.2	Supply elasticity of hotels versus peer hosts	5
3.3	Hotel outcomes and Airbnb entry	6
4	Structural model (all equations + interpretation)	6
4.1	Consumer demand	6
4.2	Hotel supply: Cournot competition with capacity-related costs	7
4.3	Peer-host supply	7
4.4	Demand estimation and survey substitution moments	7
4.5	Hotel and Airbnb cost estimation	8
5	Identification and credibility	8
5.1	What identifies demand?	8
5.2	What identifies supply?	8
5.3	Main limitations	9
6	Counterfactuals and welfare results	9
6.1	Removing Airbnb	9
6.2	Compression nights	9

6.3	Taxes, quotas, and growth	10
7	Figures and tables (original titles and original numbers from the paper)	10
7.1	Visual guide	10
7.2	Figure 1. Growth of Airbnb	13
7.3	Table 1—Descriptive Statistics	13
7.4	Figure 2. Airbnb Penetration and Hotel Revenues per Available Room	13
7.5	Table 2—City Characteristics and Size of Airbnb	13
7.6	Table 3—The Supply Elasticity of Hotels and Peer Hosts	13
7.7	Table 4—Hotel Revenues and Airbnb Entry	14
7.8	Figure 3. Prices and Occupancy Rates	14
7.9	Table 5—Estimates of Selected Demand Parameters	14
7.10	Figure 4. Estimated Utilities for Accommodation Options across Cities	14
7.11	Figure 5. Estimated Hotel Costs	14
7.12	Figure 6. Mean Costs of Airbnb Hosts in New York City	15
7.13	Table 6—Aggregate Surplus (MM)	15
8	Short summary	15
9	Conclusion	15

1 Big picture

- **Question.** What are the welfare effects of peer entry in short-term accommodation markets? In particular, how does Airbnb affect consumers, incumbent hotels, peer hosts, and local governments?
- **Core economic issue.** Airbnb is not just another hotel chain. It adds *flexible supply*: hosts have alternative uses for their rooms and can enter or exit the market depending on demand, prices, and the inconvenience of hosting. Hotels have *dedicated supply*: rooms exist for accommodation use, but capacity is costly to expand and is often fixed in the short run.
- **Main mechanism.** Peer hosts expand supply precisely when hotels are most constrained: high-demand cities, holidays, festivals, and compression nights. This makes Airbnb especially valuable when hotel capacity is close to binding.
- **Empirical setting.** The paper studies major US cities from 2011 to 2015 using proprietary Airbnb data, hotel data from STR, air-passenger demand data, Google Trends, and city-level housing/regulatory covariates.
- **Reduced-form facts.** Airbnb grows more in cities with high hotel revenues per available room, high demand growth/volatility, constrained hotel expansion, and low apparent hosting costs. Peer supply is much more elastic than hotel supply. Airbnb reduces hotel revenues mainly in hotel-constrained cities, and mainly through lower hotel prices rather than lower occupancy.
- **Structural contribution.** The paper estimates a model of differentiated demand, Cournot hotel supply with capacity-related costs, and a competitive fringe of price-taking peer hosts with heterogeneous costs.
- **Main welfare result.** In 2014, Airbnb generates about \$305 million of consumer surplus and \$112 million of peer-host surplus in the ten largest Airbnb cities. These benefits are large per transaction: about \$70 in consumer surplus and \$26 in peer-host surplus per Airbnb room-night.
- **Incumbent effect.** Hotels lose from peer entry. Without Airbnb, hotel revenues would be about 1.6 percent higher and hotel profits would rise by about \$165 million in the authors' 2014 counterfactual.
- **Policy result.** Equalizing lodging taxes between Airbnb and hotels raises local tax revenue but reduces consumer and host surplus. Quotas that allow Airbnb mostly on the highest-demand days preserve a substantial share of the gains relative to an outright ban.
- **Takeaway.** The value of peer production is not only product variety. It is also the ability to mobilize otherwise idle capacity exactly when dedicated capacity is scarce.

2 Data and measurement (key objects)

2.1 Markets, products, and sample

- **Market definition.** A market is a city-day. The short-run model treats one city on one day as the relevant competitive environment.
- **Main descriptive sample.** The paper studies the 50 largest US hotel markets between January 2011 and December 2015.

- **Structural estimation sample.** The structural model is estimated using the ten cities with large Airbnb entry: Austin, Boston, Los Angeles, Miami, New York, Oakland, Portland, San Francisco, San Jose, and Seattle.
- **Hotel products.** Hotels are grouped into six quality tiers: luxury, upper-upscale, upscale, upper-midscale, midscale, and economy.
- **Airbnb products.** Airbnb listings are grouped into four quality types: Airbnb luxury, Airbnb upscale, Airbnb midscale, and Airbnb economy.

Interpretation. The product structure is coarse but useful. It lets the model distinguish substitution between hotels and Airbnb, while still keeping the supply side tractable.

2.2 Data sources

- **Airbnb data.** Proprietary Airbnb data provide listings, active listings, available listings, bookings, prices paid by travelers, host receipts, commissions, and lodging-tax information at aggregated market-product levels.
- **Hotel data.** STR provides city-day-tier measures of hotel rooms available, rooms sold, and room revenues.
- **Demand shifters.** Sabre air-travel data measure incoming travelers. Google Trends measures searches for hotels in each city.
- **City characteristics.** Saiz’s undevelopable-land measure and the Wharton Residential Land Use Regulatory Index proxy for constraints on expanding hotel/housing supply. Census demographics proxy for peer hosts’ costs of hosting.
- **Survey moment.** A survey of Airbnb travelers asks what they would have done if Airbnb had not been available. This helps identify substitution between Airbnb and hotels.

Interpretation. The data strategy combines platform data, hotel industry data, and external demand instruments. The survey moment is especially important because observed market shares alone cannot fully reveal whether Airbnb consumers would switch to hotels or leave the market.

2.3 Main empirical objects

- **Airbnb supply share:**

$$\text{Airbnb share}_{mt} = \frac{\text{available Airbnb rooms}_{mt}}{\text{available Airbnb rooms}_{mt} + \text{hotel rooms}_{mt}}.$$

- **Occupancy:** rooms sold divided by rooms available.
- **Price:** transacted price per room-night.
- **Hotel RevPAR:** hotel revenue per available room.
- **Active listings versus available listings:** active listings are used as an instrument for available listings because hosts may endogenously change availability in response to demand shocks.

Key distinction. Airbnb “available listings” are closer to actual supply but potentially endogenous. Airbnb “active listings” are less responsive to day-level demand shocks and therefore useful as an instrument.

3 Reduced-form evidence (all regressions + interpretation)

3.1 City-level predictors of Airbnb penetration

The first empirical specification relates Airbnb penetration in city m to hotel capacity constraints, demand conditions, and demographic proxies for hosting costs:

$$\begin{aligned} \text{share_airbnb}_m = & \alpha_1 \text{Saiz}_m + \alpha_2 \text{WRLURI}_m + \alpha_3 \text{share_child}_m + \alpha_4 \text{share_unmarried}_m \\ & + \alpha_5 \text{airpass_sd}_m + \alpha_6 \text{airpass_growth}_m + \alpha_7 \log(\text{RevPAR}_m) + \alpha_8 \log(\text{market size}_m) + \varepsilon_m. \end{aligned} \quad (1)$$

Read. The left-hand side is Airbnb’s share of available rooms in the last quarter of 2015. The regressors measure constraints on hotel expansion, demand variability, demand growth, hotel revenue per room, and demographics that affect the cost of hosting.

Interpretation. Airbnb grows where hotel rooms are expensive/scarc, where travel demand is volatile or growing, and where residents are more likely to find hosting feasible. This supports the model’s core idea: peer supply enters where flexible capacity has high value.

3.2 Supply elasticity of hotels versus peer hosts

The paper estimates the short-run supply elasticity separately for hotels and Airbnb:

$$\log(Q_{mt}) = \chi \log(K_{mt}) + \kappa \log(p_{mt}) + \mu_{mt} + \varepsilon_{mt}, \quad (2)$$

where Q_{mt} is bookings, K_{mt} is available capacity, p_{mt} is average transacted price, and μ_{mt} includes city, seasonality, and day-of-week controls.

Identification problem. Prices and quantities are jointly determined. Higher prices may reflect demand shocks rather than supply responses.

Instruments.

- Hotel prices are instrumented with incoming air passengers and lagged Google Trends.
- Airbnb prices and available listings are instrumented with demand shifters and active listings.

Result. Table 3 shows that the hotel price elasticity is about 1.29, while the Airbnb price elasticity is about 3.89. Peer supply is roughly three times as elastic as hotel supply.

Interpretation. Hotels cannot easily change capacity in the short run. Peer hosts can decide whether to make rooms available when prices are high. This flexibility is central to the welfare gains from Airbnb.

3.3 Hotel outcomes and Airbnb entry

The paper then estimates the competitive effect of Airbnb on hotel outcomes:

$$y_{mt} = \alpha_1 \log(\text{Airbnb}_{mt}) + \alpha_2 \log(\text{Airbnb}_{mt}) \times \text{constrained}_m \quad (2)$$

$$+ \beta_1 \log(\text{hotelrooms}_{mt}) + \beta_2 \log(\text{hotelrooms}_{mt}) \times \text{constrained}_m \quad (3)$$

$$+ \gamma \log(\text{gtrend}_{mt}) + \delta \log(\text{travelers}_{mt}) + \theta_{mt} + \nu_{mt}. \quad (3)$$

Here y_{mt} is one of three hotel outcomes: log RevPAR, occupancy rate, or log price. The constrained-city dummy equals one when housing supply elasticity is below the median.

Read. α_1 measures the effect of Airbnb in relatively unconstrained cities. α_2 measures the extra effect in constrained cities.

Result. A 10 percent increase in available Airbnb listings lowers hotel revenue per available room by about 0.57 percent more in constrained cities than in unconstrained cities. The effect is mainly on hotel prices rather than occupancy.

Interpretation. When hotels are capacity constrained, Airbnb cannot mainly steal large quantities of hotel rooms—hotels are already full. Instead, Airbnb reduces hotel pricing power at the peak.

4 Structural model (all equations + interpretation)

4.1 Consumer demand

Consumers choose among hotel tiers, Airbnb listing types, and an outside option. Utility from option j in market n is

$$u_{ijn} = \mu_{ijn} + \alpha_i(1 + \tau_{jn})p_{jn} + \epsilon_{ijn}. \quad (4)$$

- μ_{ijn} is mean utility plus consumer heterogeneity.
- p_{jn} is the supplier price.
- τ_{jn} is the wedge between the traveler price and supplier receipt, including taxes and platform fees.
- α_i is price sensitivity.
- ϵ_{ijn} is an extreme-value taste shock.

Given the random-coefficients logit structure, demand for accommodation type j is

$$Q_{jn}(p_{jn}, p_{-jn}) = D_n \int \frac{\exp\{\mu_{ijn} + \alpha_i(1 + \tau_{jn})p_{jn}\}}{1 + \sum_{j'} \exp\{\mu_{ij'n} + \alpha_i(1 + \tau_{j'n})p_{j'n}\}} dH(i). \quad (5)$$

Plain English. Travelers differ in their value for hotels versus Airbnb and in their price sensitivity. The model can therefore ask whether Airbnb bookings are market-expanding or mostly diverted from hotels.

4.2 Hotel supply: Cournot competition with capacity-related costs

Hotels choose quantities. Within a tier, hotels are treated as homogeneous. Let Q_{hn} be aggregate tier-level quantity, $q_{hn} = Q_{hn}/N_{hn}$ be a representative hotel's quantity, and k_{hn} be hotel-level capacity.

The hotel maximizes

$$\max_{q_{hn}} q_{hn} p_{hn}(Q_{hn}, Q_{-hn}, Q_{an}) - q_{hn} c_{hn} - \frac{\gamma_{hn}}{2} \mathbf{1}\{q_{hn} > \nu k_{hn}\} (q_{hn} - \nu k_{hn})^2, \quad (4)$$

where $\nu = 0.85$ in the paper.

The first-order condition can be written as

$$p_{hn} = -\frac{1}{N_{hn}} \frac{Q_{hn}}{Q'_{hn}} + c_{hn} + \gamma_{hn} \mathbf{1}\{q_{hn} > \nu k_{hn}\} (q_{hn} - \nu k_{hn}). \quad (6)$$

Plain English. Hotel prices rise as rooms become scarce. The term with γ_{hn} captures the sharply increasing marginal cost/pricing pressure when hotels approach capacity.

4.3 Peer-host supply

Airbnb hosts are modeled as a competitive fringe. Hosts of type a draw marginal costs from a normal distribution with mean ω_{an} and standard deviation σ_{an} . They host only if price exceeds cost:

$$Q_{an}(p_{an}, p_{-an}, p_{hn}) = K_{an} \mathbb{P}(c \leq p_{an}) = K_{an} \Phi\left(\frac{p_{an} - \omega_{an}}{\sigma_{an}}\right). \quad (7)$$

Plain English. Higher Airbnb prices draw more hosts into the market. Heterogeneous host costs generate an upward-sloping but flexible peer-supply curve.

4.4 Demand estimation and survey substitution moments

Demand is estimated using two sets of moments.

The standard BLP market-share moment is

$$m_{1,jn} = (\delta_{jn} - \hat{\delta}_{jn}) Z_{jn}, \quad (8)$$

where δ_{jn} rationalizes observed market shares, $\hat{\delta}_{jn}$ is the model-predicted mean utility, and Z_{jn} are instruments or controls.

The second moment uses survey evidence on what Airbnb travelers would have chosen if Airbnb were unavailable. In compact form, the model is disciplined to match

$$m_2 = \text{model-implied Airbnb-to-hotel substitution share} - \text{survey-implied Airbnb-to-hotel substitution share}. \quad (9)$$

Why this matters. Without the survey moment, the model could fit observed shares but still mismeasure whether Airbnb is mostly stealing hotel guests or expanding the market.

4.5 Hotel and Airbnb cost estimation

After estimating demand and price derivatives, the hotel supply equation becomes

$$p_{hn} + \frac{1}{N_{hn}} \frac{Q_{hn}}{Q'_{hn}} = \theta X_{hn} + \gamma_{hn} \mathbf{1}\{q_{hn} > \nu k_{hn}\} (q_{hn} - \nu k_{hn}) + \varepsilon_{hn}. \quad (10)$$

For Airbnb, equation (7) implies

$$\Phi^{-1} \left(\frac{Q_{an}}{K_{an}} \right) = \beta_a p_{an} + \gamma_a X_{an} + \varepsilon_{an}. \quad (11)$$

The cost parameters are recovered from

$$\sigma_{an} = \frac{1}{\beta_a}, \quad \omega_{an} = \frac{\gamma_a X_{an} + \varepsilon_{an}}{\beta_a}.$$

Interpretation. The structural model turns observed quantities, prices, market shares, and survey substitution into primitives: demand utilities, hotel costs, peer-host cost distributions, and capacity-related pricing pressure.

5 Identification and credibility

5.1 What identifies demand?

- **Market shares** discipline mean utilities for hotels and Airbnb options.
- **Price variation** across days and cities disciplines price sensitivity.
- **Survey substitution** helps pin down whether Airbnb users would substitute to hotels or leave the market if Airbnb disappeared.
- **Product tiers** allow substitution patterns to differ between luxury, midscale, and economy options.

Interpretation. The key identification problem is substitution. Airbnb may look small in market share, but if most Airbnb travelers would not have used hotels, its welfare effect is very different from a simple hotel-diversion model.

5.2 What identifies supply?

- **Hotel supply** is identified from hotel prices, quantities, demand elasticities, and capacity-related price increases.
- **Peer-host supply** is identified from how Airbnb bookings and active/available listings respond to prices and demand shifters.

- **Instruments** include incoming air passengers, lagged Google Trends, and active Airbnb listings.

Interpretation. The supply side needs exogenous demand shifts to trace supply responses rather than demand movements. The central empirical fact is that Airbnb supply responds much more elastically than hotel supply.

5.3 Main limitations

- The model is short-run and does not fully capture hotel investment responses or long-run entry/exit.
- Airbnb listings are aggregated into broad quality groups; within-city location heterogeneity is only partly addressed.
- The model does not fully incorporate housing-market externalities, neighborhood effects, or distributional consequences for renters and residents.
- Welfare calculations depend on the assumed market size and on substitution patterns between Airbnb, hotels, and the outside option.

Bottom line. The paper is strongest on short-run accommodation-market welfare: consumers, hotels, hosts, and lodging taxes. It is less complete as a full urban welfare analysis because it does not fully model housing-market spillovers.

6 Counterfactuals and welfare results

6.1 Removing Airbnb

The key counterfactual removes peer supply and recomputes market outcomes.

- **Consumer surplus loss:** about \$305 million in 2014.
- **Peer-host surplus loss:** about \$112 million, because peer hosts disappear.
- **Hotel gains:** hotel revenues rise from \$26.803 billion to \$27.238 billion and profits rise from \$5.687 billion to \$5.852 billion.
- **Per Airbnb room-night:** consumer surplus is about \$70 and peer-host surplus is about \$26.

Interpretation. Airbnb creates more surplus for consumers and hosts than the profit it takes away from hotels. The welfare gains are large for the participants even though Airbnb is small relative to total accommodation-market revenue.

6.2 Compression nights

Compression nights are nights when at least one hotel tier reaches 95 percent occupancy. They represent 19.6 percent of markets in 2014.

- The no-Airbnb consumer surplus loss on compression nights is \$121 million.

- This is roughly 40 percent of the total consumer surplus loss from removing Airbnb.
- During high-demand periods, Airbnb bookings are especially market-expanding: many Airbnb travelers would not switch to hotels if Airbnb were unavailable.

Interpretation. The value of peer supply is most visible when hotels run into physical capacity constraints.

6.3 Taxes, quotas, and growth

- **Lodging-tax parity.** Applying hotel lodging taxes to Airbnb reduces consumer and host surplus but increases government tax revenue by about \$72 million.
- **Quotas.** Allowing Airbnb only on the 90 highest-demand days preserves a meaningful share of consumer benefits relative to an outright ban, because benefits are concentrated at demand peaks.
- **Double Airbnb rooms.** Doubling active Airbnb listings increases consumer surplus by about \$130 million and peer-host surplus by about \$34 million, while further reducing hotel revenues and profits.

Policy takeaway. Policies that preserve flexible supply during peak demand retain much of Airbnb’s efficiency value. Blanket bans destroy surplus that arises precisely because peer supply appears when hotel capacity is scarce.

7 Figures and tables (original titles and original numbers from the paper)

The paper contains six figures and six tables. Below, I keep the paper’s original figure/table numbers and titles. Adjacent figures/tables are placed two side-by-side.

7.1 Visual guide

Figure 1. Growth of Airbnb

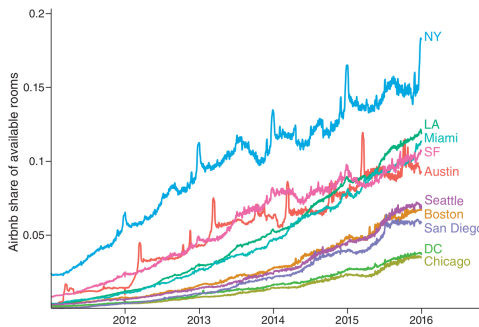


Table 1—Descriptive Statistics

	N	Mean	SD	25 pct	Median	75 pct
Mean hotel occupancy	50	0.67	0.07	0.62	0.66	0.72
Standard deviation hotel occupancy	50	0.14	0.03	0.12	0.14	0.15
Mean hotel price in \$	50	110.77	35.66	88.45	100.90	124.29
Standard deviation hotel price	50	17.97	9.84	10.42	14.98	23.22
Mean hotel revenue (\$ thousands)	50	3,945	3,697	1,639	2,630	5,013
Mean Airbnb occupancy	50	0.16	0.05	0.13	0.14	0.20
Standard deviation Airbnb occupancy	50	0.10	0.02	0.08	0.10	0.11
Mean Airbnb price in \$	50	113.93	25.80	97.76	103.77	125.78
Standard deviation Airbnb price	50	32.09	12.43	23.18	29.41	38.66
Mean Airbnb to hotel price ratio	50	1.08	0.28	0.94	1.03	1.16
Standard deviation price ratio	50	0.32	0.16	0.21	0.29	0.37
Airbnb share of available rooms	50	0.04	0.03	0.01	0.02	0.06
Airbnb share of potential guests	50	0.05	0.04	0.01	0.03	0.08
Airbnb share of housing units	50	0.001	0.001	0.0004	0.001	0.002

Notes: This table shows hotel and Airbnb descriptive statistics for the 50 cities in our sample. For each city, we compute the mean and standard deviation of daily metrics for hotels and Airbnb listings between January 2011 and December 2015. The metrics we consider are occupancy rates, prices per room-night, revenues, ratio of Airbnb to hotel prices. The last three rows show Airbnb size as a share of available rooms, potential guests, and housing units in the last quarter of our sample, October–December 2015. The Airbnb share of available rooms is computed as the average daily share of available rooms (Airbnb listings divided by the sum of Airbnb listings and hotel rooms). The Airbnb share of potential guests is computed as the average share of available rooms adjusted for their realized capacity, i.e., number of guests occupying a room. To make this adjustment, we have data on Airbnb realized number of guests per room at the city-day-listing type level. Since we do not have the same metric for hotels, we assume that the typical hotel has the same number of average guests as a *Midscale* Airbnb listing in the same city. The Airbnb share of potential guests is typically higher than the Airbnb share of available rooms because an Airbnb listing is on average occupied by more guests than hotel rooms. Finally, the Airbnb share of housing units is the average of the ratio of available Airbnb listings divided by the number of housing units in the metropolitan statistical area.

Figure 2. Airbnb Penetration and Hotel Revenues per Available Room

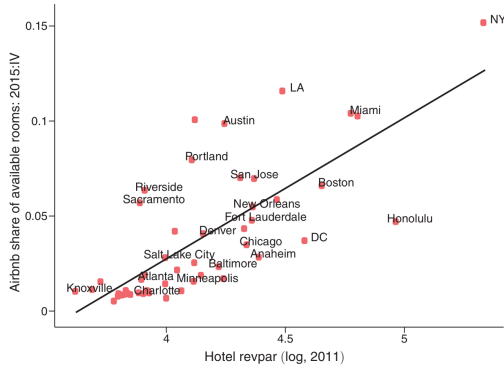


Table 2—City Characteristics and Size of Airbnb

	Airbnb share of rooms, 2015:IV	
	(1)	(2)
Undevelopable area	0.036 (0.017)	0.026 (0.015)
Wharton residential land use index (WRLURI)	0.009 (0.004)	0.005 (0.004)
SD. incoming air passengers, 2011	0.002 (0.001)	0.00003 (0.001)
Percent growth in air passengers, 2012–2011	0.125 (0.067)	0.107 (0.058)
Percent never married	0.504 (0.164)	0.308 (0.152)
Percent children	−0.399 (0.203)	−0.218 (0.182)
Log(revenue per room, 2011)		0.061 (0.017)
Log(market size)	−0.008 (0.008)	−0.012 (0.007)
Constant	0.052 (0.091)	−0.133 (0.093)
Observations	46	46
R ²	0.668	0.758

Notes: This table shows linear regressions of the size of Airbnb on market characteristics linked to supply constraints, demand volatility, and the costs of hosting (equation (1)). The size of Airbnb is the average daily share of rooms in the last quarter of 2015. The standard deviation of incoming air passengers is divided by 10,000 to make the coefficient comparable to the other variables. Descriptive statistics are shown in Table 1 (for the outcome variable) and online Appendix Table E1 (for the predictors). Market size is measured as the average number of hotel rooms and Airbnb listings available in the last quarter of 2015.

Table 3—The Supply Elasticity of Hotels and Peer Hosts

	log(hotel rooms booked + 1) (1)	log(Airbnb rooms booked + 1) (2)
log(hotel rooms + 1)	0.543 (0.215)	
log(hotel price)	1.289 (0.103)	
log(Airbnb available listings + 1)		0.385 (0.142)
log(Airbnb price)		3.893 (0.288)
IV	Yes	Yes
City fixed effects	Yes	Yes
Year-month fixed effects	Yes	Yes
Day of week fixed effects	Yes	Yes
Observations	90,900	84,959
R ²	0.954	0.774

Notes: The table shows results of IV regressions of the log of hotel and Airbnb bookings on the corresponding price and room availability (equation (2)). In column 1 we instrument for hotel prices with the one week lag of the log of the Google search trends and the log of arriving (not returning) flight travelers. In column 2 we instrument for both Airbnb prices and the number of available listings. We use three instruments: the one week lag of the log of the Google search trends and the log of arriving flight travelers as in column 1, plus the number of Airbnb active listings. First stage results are reported in online Appendix Table E2 and OLS results are reported in online Appendix Table E3. Standard errors are clustered at the city level. Adding the city-day observations with no Airbnb bookings (and using hotel prices in column 2) does not change the results. Results would not change if in column 2 we included the log of departing air travelers and the one-week lag of the log of local Google Search Trends for hotels outside of the city as additional controls.

Table 4—Hotel Revenues and Airbnb Entry

	log(revpar) (1)	Occupancy rate (2)	log(price) (3)
log(incoming air passengers)	1.104 (0.063)	0.371 (0.041)	0.482 (0.040)
log(Google search trend)	0.246 (0.042)	0.076 (0.012)	0.109 (0.024)
log(hotel rooms + 1)	−0.936 (0.326)	−0.521 (0.137)	−0.089 (0.168)
log(hotel rooms + 1) × Inelastic housing supply	−0.475 (0.370)	0.055 (0.174)	−0.612 (0.281)
log(Airbnb available listings + 1)	0.020 (0.016)	−0.002 (0.007)	0.021 (0.010)
log(Airbnb available listings + 1) × Inelastic housing supply	−0.057 (0.025)	−0.002 (0.010)	−0.054 (0.022)
IV	Yes	Yes	Yes
City fixed effects	Yes	Yes	Yes
Year-quarter fixed effects	Yes	Yes	Yes
Day of week fixed effects	Yes	Yes	Yes
Observations	90,900	90,900	90,900
R ²	0.740	0.591	0.856

Notes: This table shows results of IV estimates of equation (3), where the size of Airbnb is measured as the number of available listings. The Google search trend is a one-week lag. The instruments for available listings and its interaction with the dummy for inelastic housing supply are the number of active listings and its corresponding interaction with the dummy. The dependent variable is log revenue per available room in column 1, occupancy rate in column 2, and log price in column 3. First stage results are reported in online Appendix Table E4 and OLS results are reported in online Appendix Table E5. Results for different hotel tiers are presented in online Appendix Table E6, and results using different measures of the size of Airbnb are in online Appendix Table E7. Standard errors are clustered at the city level.

Figure 3. Prices and Occupancy Rates

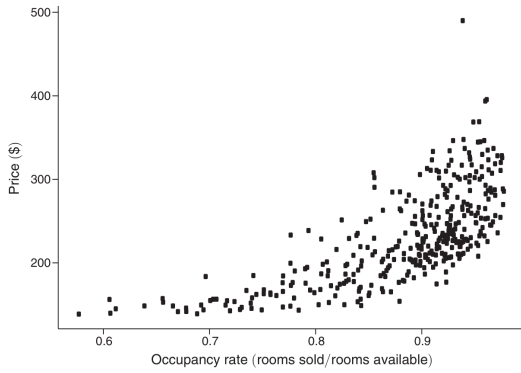


Figure 4. Estimated Utilities for Accommodation Options across Cities

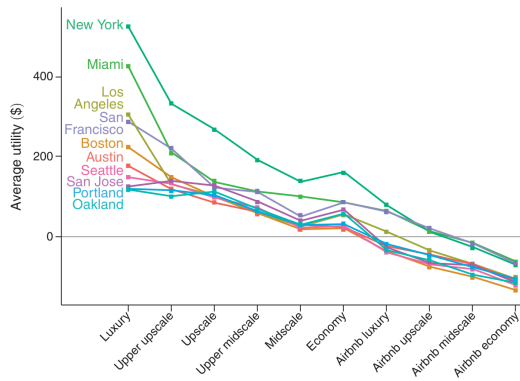


FIGURE 4. ESTIMATED UTILITIES FOR ACCOMMODATION OPTIONS ACROSS CITIES

Figure 6. Mean Costs of Airbnb Hosts in New York City

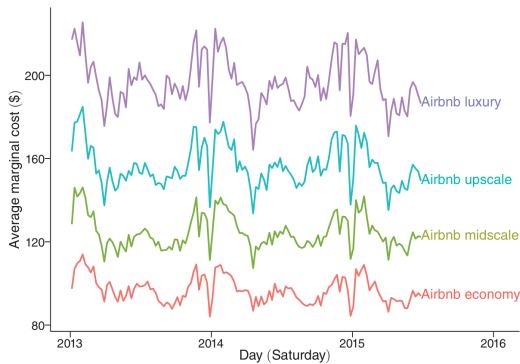


Table 5—Estimates of Selected Demand Parameters

Parameter	Random coefficients logit		Standard logit	
	Estimate	Std. error	Estimate	Std. error
log Google trend	2.355	0.281	1.783	0.059
Price	−0.031	0.002	−0.025	0.001
Std. deviation on inside option	1.725	1.060	.	.
Std. deviation on price	0.004	0.004	.	.

Note: This table displays the estimates and standard errors for selected parameters in travelers' utility (equation (5)).

Figure 5. Estimated Hotel Costs

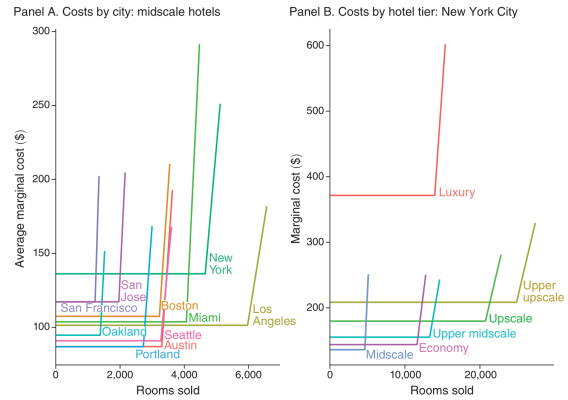


Table 6—Aggregate Surplus (MM)

	Consumers	Hotels			Peer hosts			Government
	Change in consumer surplus	Rooms sold	Revenues	Profits	Rooms sold	Revenues	Peer surplus	Lodging taxes
<i>Panel A. All markets in 2014</i>								
Baseline		146	26,803	5,687	4.38	517	112	3,986
No Airbnb (unconstrained)	−147	149	27,412	5,833				4,071
No Airbnb	−305	148	27,238	5,852				4,045
Airbnb with lodging tax	−65	146	26,891	5,718	3.47	377	82	4,058
Airbnb with quotas	−157	147	27,106	5,754	1.49	181	40	4,027
Double Airbnb rooms	130	145	26,630	5,623	6.18	672	146	3,962
<i>Panel B. Compression nights in 2014 (19.6 percent of all markets)</i>								
Baseline		33	7,240	3,414	1.16	145	31	1,084
No Airbnb (unconstrained)	−39	33	7,435	3,505				1,111
No Airbnb	−121	33	7,341	3,495				1,097
Airbnb with lodging tax	−24	33	7,258	3,429	0.94	107	23	1,103
Airbnb with quotas	−14	33	7,256	3,424	1.01	125	28	1,086
Double Airbnb rooms	53	33	7,199	3,380	1.65	190	41	1,078

Notes: This table displays the outcomes for consumers, hotels, peer hosts, and local governments under the baseline scenario and five alternative scenarios: two scenarios without Airbnb and three scenarios with Airbnb and regulation. “Unconstrained” refers to the counterfactual scenario in which Airbnb options do not exist, hotels do not adjust prices and can accommodate any additional bookings regardless of their actual capacity. In the “No Airbnb” counterfactual, we let hotel prices readjust in response to the absence of Airbnb and accounting for hotel capacity constraints. The next counterfactuals consider new equilibrium prices and quantities under different regulation. The “Airbnb with lodging tax” counterfactual keeps Airbnb availability at the baseline level, but Airbnb travelers are charged the same lodging tax rate as hotel travelers. The “Airbnb with quotas” counterfactual allows for Airbnb rooms to exist during the 90 days in a year with the largest number of travelers choosing to book accommodations in a particular city in the baseline scenario. For the other 275 days, Airbnb rooms are not allowed and so the equilibrium prices and quantities mirror those in the “No Airbnb” counterfactual. Note that those 90 days when Airbnb is allowed are not the same across all cities, but rather are determined independently for each city. The “Double Airbnb Rooms” counterfactual doubles the number of active Airbnb listings. Panel A displays metrics aggregated across all cities and nights in 2014 while panel B focuses on compression nights, i.e., the markets when at least one hotel tier has an occupancy rate of 95 percent or more (19.6 percent of markets in 2014 are considered compression nights). All variables are in millions. For heterogeneity of the effects across cities, see online Appendix Tables E16 through E18. For counterfactuals without consumer heterogeneity, see online Appendix Table E20.

7.2 Figure 1. Growth of Airbnb

Read. The figure plots Airbnb’s share of available accommodation rooms over time for ten large Airbnb cities.

Key fact. Airbnb grows rapidly but heterogeneously. New York grows fastest and shows sharp temporary spikes around high-demand dates such as New Year’s Eve.

Why it matters. The figure motivates the flexible-supply mechanism: peer hosts expand participation when market conditions make hosting attractive.

7.3 Table 1—Descriptive Statistics

Read. The table summarizes hotel and Airbnb prices, occupancy rates, revenue measures, and Airbnb size across the 50 cities.

Key numbers. Average hotel price is about \$111, average Airbnb price is about \$114, hotel occupancy is about 67 percent, and Airbnb occupancy is about 16 percent. Airbnb’s average share of available rooms is about 4 percent in the last quarter of 2015.

Why it matters. Airbnb is small relative to hotels but not negligible, and it has much lower occupancy. That low occupancy is consistent with flexible supply and high host cost heterogeneity.

7.4 Figure 2. Airbnb Penetration and Hotel Revenues per Available Room

Read. The figure plots Airbnb penetration against hotel RevPAR in 2011.

Key fact. Airbnb penetration is positively related to hotel revenue per available room.

Why it matters. Peer entry is strongest where hotels are expensive and capacity is valuable. This supports the idea that peer supply enters where dedicated supply is scarce.

7.5 Table 2—City Characteristics and Size of Airbnb

Read. The table regresses Airbnb’s 2015 city-level supply share on measures of hotel expansion constraints, demand volatility/growth, and demographic hosting-cost proxies.

Key facts. Undevelopable land, land-use regulation, demand volatility, air-passenger growth, and a larger unmarried population predict more Airbnb entry. A higher share of children predicts less Airbnb entry.

Why it matters. The coefficients line up with the model: peer entry is high when hotel supply is constrained and when hosts have lower costs of participating.

7.6 Table 3—The Supply Elasticity of Hotels and Peer Hosts

Read. The table reports IV estimates of supply responses to price.

Key numbers. The hotel supply elasticity is about 1.29. The Airbnb supply elasticity is about 3.89.

Why it matters. Peer supply is around three times as elastic as hotel supply, validating the paper’s flexible-versus-dedicated supply distinction.

7.7 Table 4—Hotel Revenues and Airbnb Entry

Read. The table estimates how available Airbnb listings affect hotel RevPAR, occupancy, and prices, with separate effects for constrained cities.

Key number. In constrained cities, a 10 percent increase in Airbnb available listings lowers hotel RevPAR by about 0.57 percent relative to unconstrained cities.

Why it matters. The competitive effect is strongest where hotel capacity is inelastic and mostly operates through prices, not quantities.

7.8 Figure 3. Prices and Occupancy Rates

Read. The figure plots prices and occupancy rates for upscale hotels in New York City in 2014.

Key fact. Prices rise steeply when occupancy gets high.

Why it matters. This justifies the structural assumption that marginal costs or pricing pressure increase sharply when hotels approach capacity.

7.9 Table 5—Estimates of Selected Demand Parameters

Read. The table reports selected demand estimates from the random-coefficients logit and standard logit models.

Key facts. The price coefficient is negative, Google search trends raise demand, and the random-coefficients model allows heterogeneity in inside-option tastes and price sensitivity.

Why it matters. The demand estimates are the foundation for consumer surplus and substitution calculations.

7.10 Figure 4. Estimated Utilities for Accommodation Options across Cities

Read. The figure plots estimated mean willingness to pay for hotel and Airbnb options across ten cities.

Key fact. Hotels generally have higher estimated mean utility than Airbnb options, but Airbnb still provides differentiated value for some travelers.

Why it matters. Airbnb creates welfare partly through variety, not only through lower hotel prices.

7.11 Figure 5. Estimated Hotel Costs

Read. The figure shows estimated hotel marginal cost curves by city and by hotel tier.

Key fact. Marginal costs/pricing pressure rise sharply as rooms sold approach capacity, especially for high-quality tiers and high-demand cities.

Why it matters. Capacity scarcity is the reason Airbnb matters most during compression periods.

7.12 Figure 6. Mean Costs of Airbnb Hosts in New York City

Read. The figure plots estimated mean costs of Airbnb hosts in New York City by listing quality over time.

Key facts. Costs differ by quality tier and vary over time. Higher-quality Airbnb listings have higher costs.

Why it matters. Cost heterogeneity explains why not all active listings are booked and why Airbnb supply can expand when prices rise.

7.13 Table 6—Aggregate Surplus (MM)

Read. The table reports welfare outcomes under the baseline and counterfactual scenarios: no Airbnb, lodging-tax parity, quotas, and double Airbnb rooms.

Key numbers. Removing Airbnb reduces consumer surplus by \$305 million and eliminates \$112 million of peer-host surplus. It raises hotel revenues and profits. Lodging-tax parity raises local tax revenue by about \$72 million.

Why it matters. Table 6 is the paper’s central welfare table. It shows that Airbnb creates surplus for consumers and hosts, reduces hotel profits, and has policy-relevant tax and quota tradeoffs.

8 Short summary

- Airbnb is valuable because it provides both product variety and flexible capacity.
- Peer hosts respond much more elastically to prices than hotels do.
- Airbnb’s effects are concentrated in cities and dates where hotels are capacity constrained.
- Removing Airbnb would reduce 2014 consumer surplus by about \$305 million and peer-host surplus by \$112 million, while increasing hotel profits.
- Regulations that keep peer supply available on high-demand days preserve much of the efficiency gain.

9 Conclusion

Core conclusion. Peer entry in accommodations creates sizable consumer and host surplus, especially when incumbent hotel capacity is scarce.

Economic intuition. Hotels provide dedicated capacity that is costly to adjust. Airbnb hosts provide flexible capacity that appears when prices and demand are high. This flexibility reduces hotel pricing power at peaks and expands market capacity.

Takeaway. The welfare value of platform-enabled peer production depends on when and where peers enter. Airbnb is most valuable not simply because it is new, but because it mobilizes supply at the moments when conventional capacity is most constrained.